# Econ 2 - Lecture 15 - 3/4/26

Lecture Quiz 8 Released today, due 3/19 @ 9:30 AM

Final Exam: Wednesday, 3/18, 9-11 AM

Today: Monetary Policy (Chapter 6), Start Agg. Demand (7.1)

Last Class: Money, Banking, The Fed, Interest Rates.

Model for M1 money:

→ What makes you want to hold onto M1?

## Demand for M1 Money

| | |
|---|---|
| M1 money = "cash" | "Bonds" |

Box of Wealth

Stocks, homes, bonds, gold } "Bonds" = "other assets"

⇒ Positive Return on assets

**Benefits of holding cash**

Buy goods/services → without transactions costs

**Cost of holding cash**

"Return on other assets/bonds"

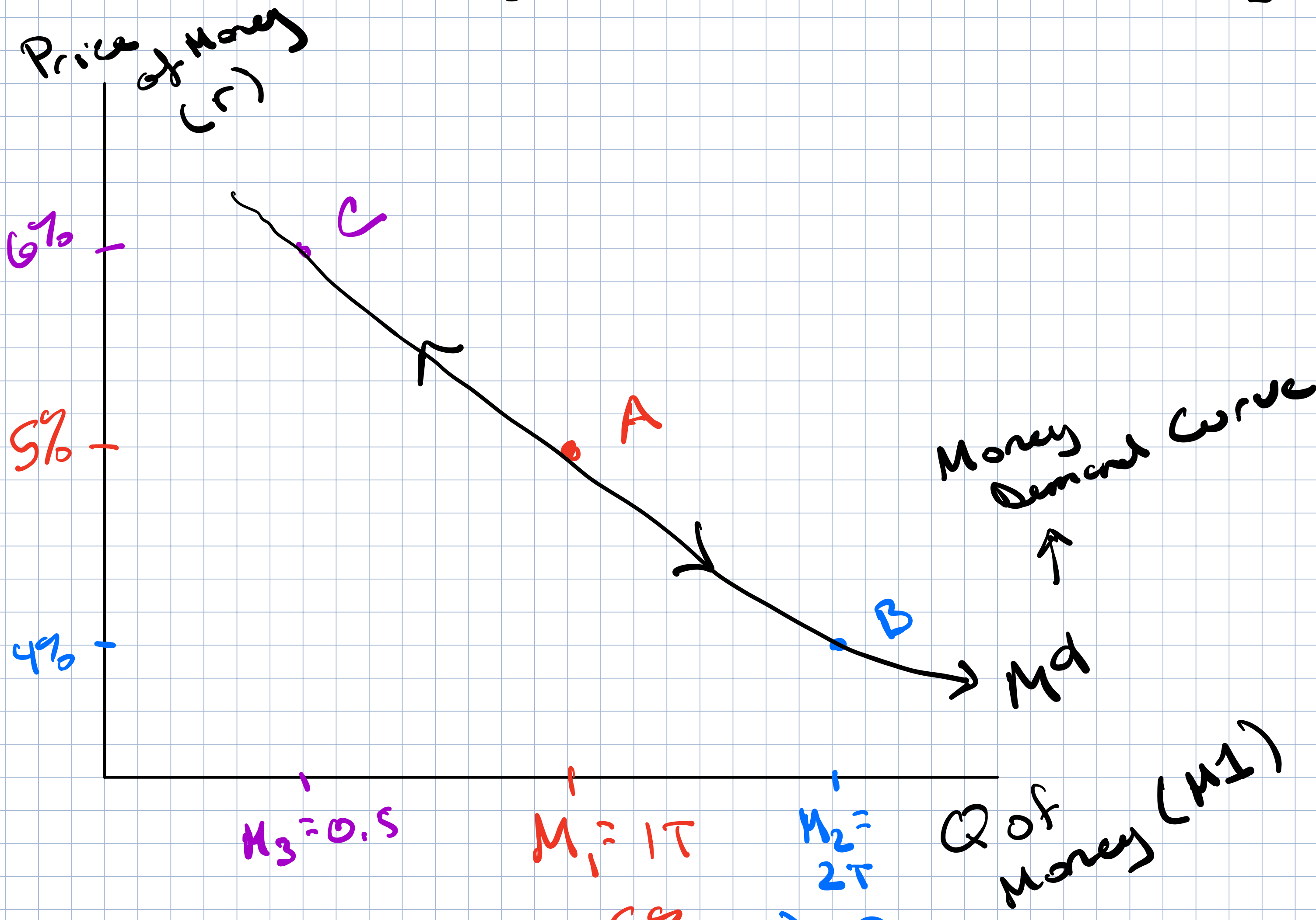
→ \$1 in cash → no return

**Opportunity Cost of Holding Cash**

Foregone interest on bonds

→ Price of money = interest rate = r

Graph the Relationship between Quantity of Money Demanded & Price of Money



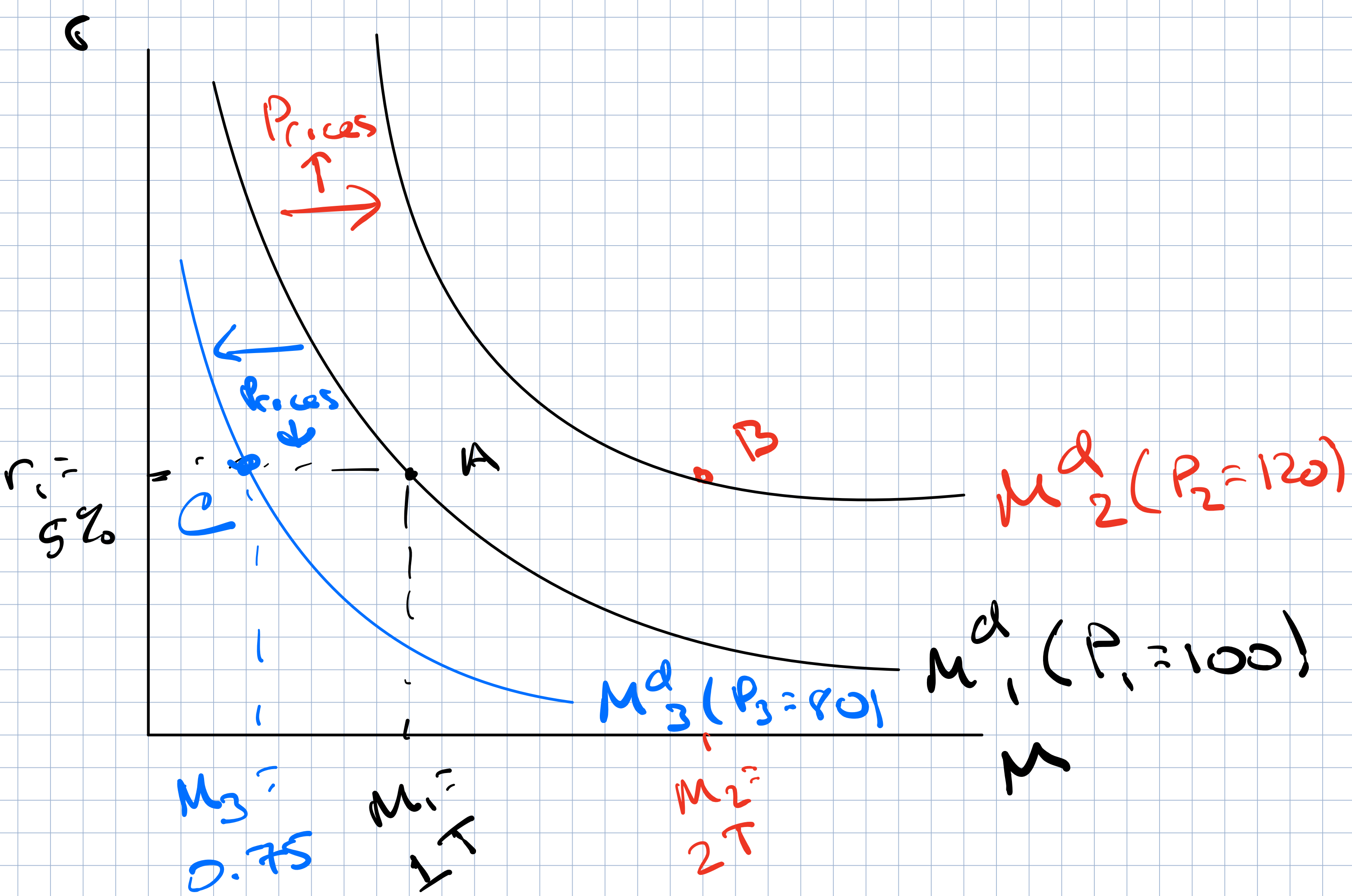

r = 5% → r = 4%


Wealth

$M^d$: As r ↓, Q of money demanded ↑
→ Move from A to B

As r ↑, Q of money demanded ↓
→ move from A to C

# What shifts $M^d$?




What changes amount of M1 you want to hold other than r?

Two Big Shifters → Primary $M^d$ shifters

1) Prices = CPI : GDP Deflator

If prices rise → require more cash

At Pt. A, $r_1 = 5\%$, Q of $M^d = 1T$, $P_1 = 100$

Prices rise to $P_2 = 120$ → At $r_1 = 5\%$, $Q = 2T$

2) Size of Wealth Box

As Wealth ↑ → Shift Right in $M^d$ ↑

Money Supply: Who determines M1 money supply?

Federal Reserve

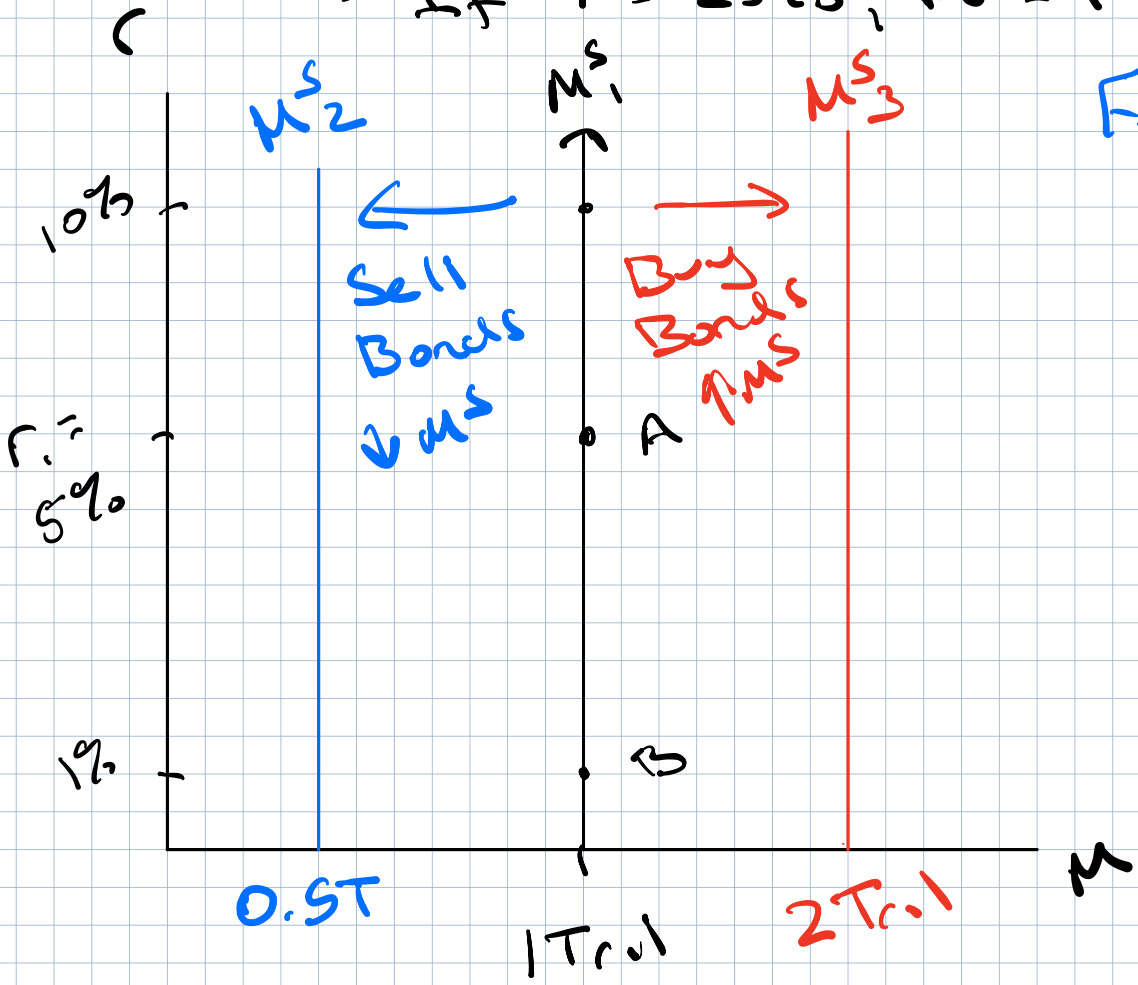
→ Open market operations

→ Set $M^s$ = 1 Trillion

↳ If r = 0%, $M^s$ = 1 Tril

↳ If r = 5%, $M^s$ = 1 Tril

↳ If r = 25%, $M^s$ = 1 Tril




Fed wants to decrease $M^s$? Sell bonds to Public

Fed wants to increase $M^s$? Buy bonds from Public

Money Supply → fixed / Perfectly inelastic

Shift the money supply curve?

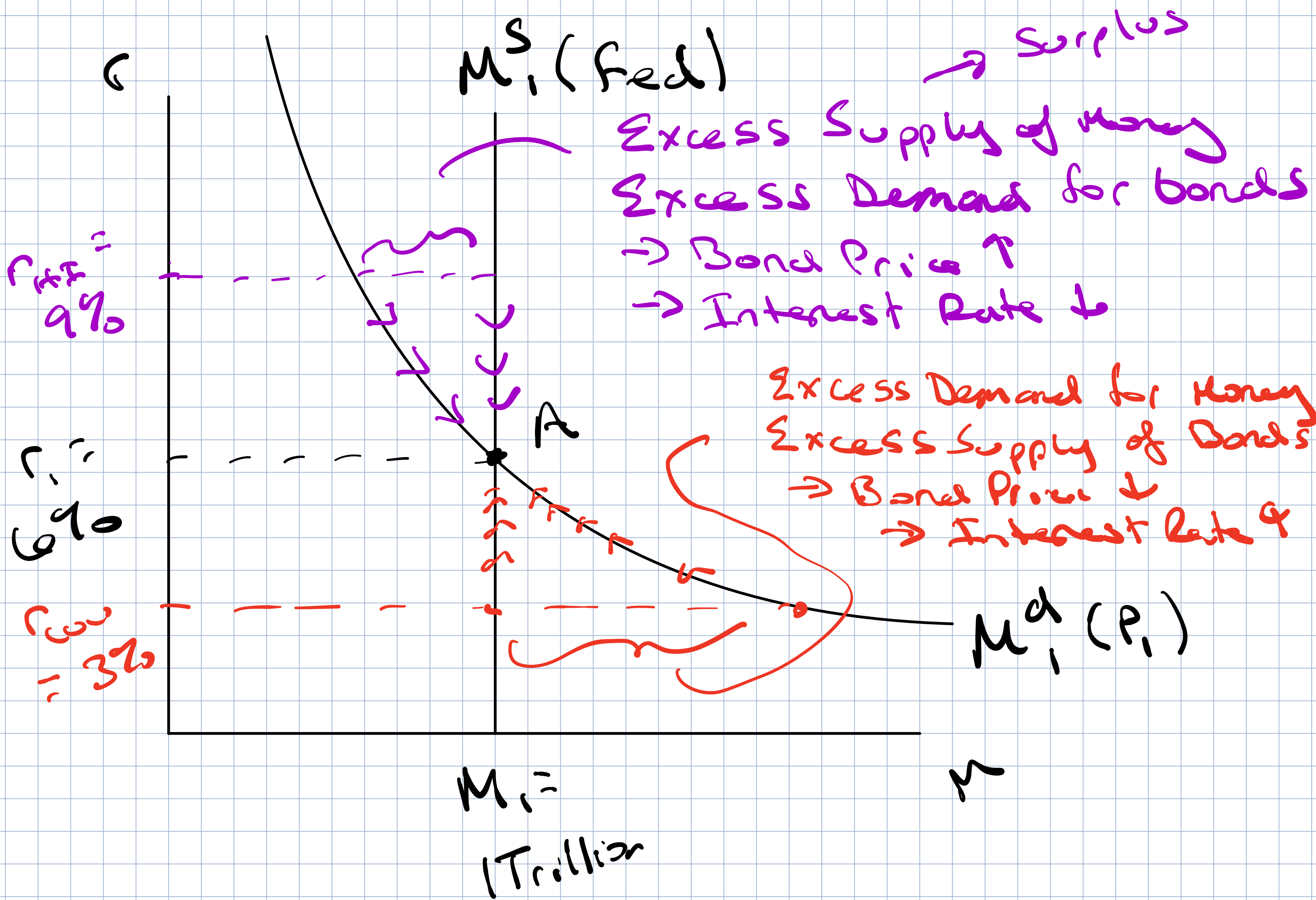
→ Buy or sell bonds to public

→ Open market purchase → Cash } held by public

→ open market sales → Cash

Combine $M^d$ & $M^s$

$M_1^d(P_1)$, $M_1^s(\text{Fed}) = 1$ Tril, $r_1 = 6\%$




Equilibrium at Pt. A

Fed set $M^s = 1$ Tril, Quantity of Money D is equal to 1 Tril when $r_1 = 6\%$

Imagine $r_{HI} > 6\%$, 9%

Price of money → too high → excess cash

→ not enough bond → Bond Price ↑, interest rate ↓

Imagine $r_{Low} < 6\%$, 3%

Price of money → too low → not enough cash

→ Too many bonds → Bond Price ↓, interest rate ↑

As bond price ↑ → Int. Rate ↓

$100, one-year bond

Bond Value = $100

Bond Price Today < $100

Price = $80

Return on Bond = Value − Price

= 100 − 80 = $20

$$\text{Interest Rate} = \frac{\text{Return on Bond}}{\text{Price of Bond}} \times 100$$

$$= \frac{20}{80} \times 100 = 25\%$$

Great Return! Excess Demand for Bonds!
Price of bonds starts to rise!

New Price = $90

$$\text{Interest Rate} = \frac{100 - 90}{90} \times 100 = \frac{10}{90} \times 100$$

$$= 11.1\%$$

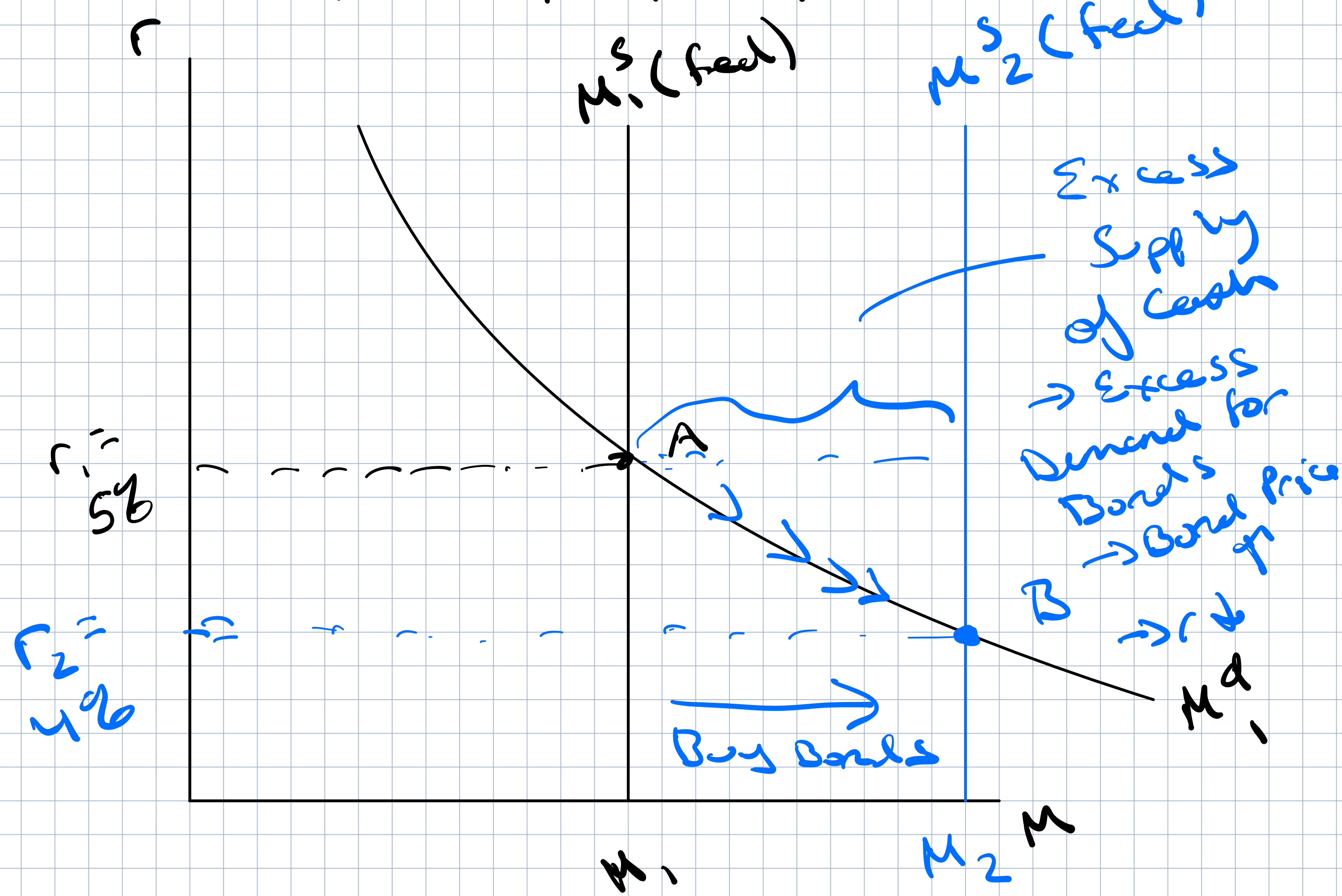
$500 bond
$475 bond price

$$\text{Interest Rate} = \frac{25}{475} \times 100$$

$$= 5.3\%$$

Connect money market to monetary policy

Start at $r_1 = 5\%$, $M^s_1 = M_1$




Fed wants to carry out open market operations → $r_1 = 5\%$ to $r_2 = 4\%$

Fed → cannot move $M^d$ curve

→ $M^s$ is movable by Fed

$\underbrace{\text{Increase } M^s}_{\text{Buy bonds}}$ from $M_1$ to $M_2$, Pt. A to Pt. B

Why does changing the interest rate matter?

In terms of AE

$$AE = C + I^P + G + NX$$

If $r_1 = 5\%$ to $r_2 = 4\%$?

1) Cheaper loans available
   → Big ticket item (car, furniture, appliance, etc.)
   → Durable Good
   → $\uparrow AC$

2) Firms that borrow → cheaper
   → $\uparrow I^P$

3) New Homes
   → $\uparrow I^P$

As $r \downarrow \Rightarrow AE \uparrow$

As $r \uparrow \Rightarrow AE \downarrow$